

1(a)	P – citrate A citric acid Q – NAD / NAD⁺ R – reduced NAD / NADH A NADH₂ S – carbon dioxide / CO₂ T – FAD U – reduced FAD / FADH₂ ;;; <i>6 correct = 3 marks</i> <i>5/4 correct = 2 marks</i> <i>3/2 correct = 1 mark</i>	3
1(b)	any two from: 1 transfer of phosphate group to ADP / ADP phosphorylated / ADP + P_i → ATP ; 2 substrate-linked phosphorylation ; A substrate-level phosphorylation R if oxidative phosphorylation 3 enzyme (catalysed reaction) ;	2

Question	Answer	Marks
1(c)	<i>any four from:</i> 1 small / water-soluble, so can move around <u>cell</u> ; 2 loss of phosphate / hydrolysis, leads to energy release ; 3 (release energy) immediately / in small packets or <i>ref.</i> 30.5 kJ (mol⁻¹) ; 4 can be, recycled / regenerated or ATP $\rightleftharpoons$ ADP + Pi ; 5 link between energy-yielding and energy-requiring reactions / AW ; 6 high turnover / described ; 7 <i>ref to</i> ATPase ;	4